

Financial Time Series Analysis

Lecture 7

- ▶ TA Session Friday April 16, 2021, 11:00am - 12:00pm (EDT)
- ▶ TA Session Monday April 19, 2021, 7:30pm - 8:30pm (EDT)

Outline

- ▶ Forecasting
 - ▶ Brief Review: AR(1)
 - ▶ MA and ARMA forecasts
 - ▶ Linear Forecasting
- ▶ Transforming from non-stationarity to stationarity
- ▶ New models (e.g. SARIMA) and packages (e.g. pmdarima)

Review: 1

- ▶ We begin with a basic result about forecasting. In the context of time series, we have data, $\mathbf{X} = (X_1, X_2, \dots, X_n)$, and we wish to forecast a future value of the series, for example X_{n+k} .
- ▶ The forecast must be made using only the available information. We symbolize that knowledge by $\mathcal{F} = \mathcal{F}(\mathbf{X})$.
- ▶ The goal is to find the best predictor that is a function of $\mathbf{X}$, that is the function must be measurable with respect to $\mathcal{F}$.
- ▶ Our criterion for “best” is the predictor that minimizes the mean square error, $E((Y - \text{predictor})^2)$.
- ▶ The answer is $E(Y|\mathcal{F})$, as $Y - E(Y|\mathcal{F})$ is uncorrelated with all $\mathcal{F}$ measurable functions.

Review: 2

- ▶ We begin with the simplest time series model, AR(1):

$$Y_t = \mu + \phi_1(Y_{t-1} - \mu) + \epsilon_t.$$

- ▶ Suppose we have data $(Y_1, \dots, Y_t)$ and we wish to forecast Y_{t+1} , the next value in the time series
- ▶ We usually assume that the noise process, ϵ_t , is $WN(0, \sigma_\epsilon^2)$. If we specialize to the case where it is an independent sequence or an MDS, the $E(\epsilon_{t+1}|\mathcal{F}_t) = 0$. So our best prediction of the noise term, ϵ_{t+1} , given the present information is 0.

Review: 3

- ▶ The AR(1) model gives $Y_{t+1} = \mu + \phi_1(Y_t - \mu) + \epsilon_{t+1}$, so the best linear forecast will be

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\phi}_1(Y_t - \hat{\mu}) + 0,$$

Note, we assume $|\phi_1| < 1$.

- ▶ To forecast Y_{t+2} , the same idea can be used, except we need to forecast both Y_{t+1} and Y_{t+2} :

$$\begin{aligned}\hat{Y}_{t+2} &= \hat{\mu} + \hat{\phi}_1(\hat{Y}_{t+1} - \mu) \\ &= \hat{\mu} + \hat{\phi}_1(\hat{\phi}_1(Y_t - \hat{\mu})) \\ &= \hat{\mu} + \hat{\phi}_1^2(Y_t - \hat{\mu}).\end{aligned}$$

In general, the k-step-ahead predictor for an AR(1) sequence is

$$\hat{Y}_{t+k} = \hat{\mu} + \hat{\phi}_1^k(Y_t - \hat{\mu}).$$

Forecasting: 1

- ▶ While we don't write out the derivation here, all the above ideas extend to AR(p) processes, except the model now contains p parameters whose estimates are used in the forecast.
- ▶ A new aspect arises when forecasting a MA(q) time series, therefore also in forecasting an ARMA(p,q) or ARIMA(p,d,q) time series model.
- ▶ To see the issue, consider a MA(1) model:

$$Y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1}.$$

Forecasting: 2

- ▶ If we wish to forecast one-step-ahead, Y_{t+1} , then we note that $Y_{t+1} = \mu + \epsilon_{t+1} + \theta_1 \epsilon_t$ and the forecast, $\hat{Y}_{t+1}$ must be based only on $\mathcal{F}_t$.
- ▶ Clearly, our best estimate of ϵ_{t+1} is 0, and we will obtain estimates of μ and θ_1 and plug them into the equation. But what do we do with ϵ_t ?
- ▶ We replace ϵ_t by the residual, $\hat{\epsilon}_t$,

$$\hat{\epsilon}_t = Y_t - \mu - \theta_1 \hat{\epsilon}_{t-1}.$$

Thus the forecast of Y_{t+1} is

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\theta}_1 \hat{\epsilon}_t.$$

Forecasting: 3

- ▶ For a MA(1) model, the autocorrelation is 0 for lags greater than 1. To forecast Y_{t+2} we note that the model gives

$$Y_{t+2} = \mu + \theta_1 \epsilon_{t+1} + \epsilon_{t+2}.$$

- ▶ Our best estimates of the two error terms, ϵ_{t+1} and ϵ_{t+2} is 0. This gives

$$\hat{Y}_{t+k} = \hat{\mu}, \quad k \geq 2.$$

- ▶ These ideas easily extend to MA(q):

$$Y_{t+1} = \mu + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \epsilon_t,$$

and

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\theta}_1 \hat{\epsilon}_{t-1} + \cdots + \hat{\theta}_q \hat{\epsilon}_{t-q} + \hat{\epsilon}_t,$$

- ▶ This carries over to ARMA(p,q) processes.

Forecasting: 4

In summary:

- ▶ We have a weakly stationary time series, $\{X_t\}$, we have observations from that time series, $(X_1, X_2, \dots, X_n)$, and we wish to predict a future value of the series, say X_{n+k} , a k -step-ahead predictor, $\hat{X}_{n+k}$.
- ▶ The criterion we are trying to optimize is minimum mean square error, $E((X_{n+k} - \hat{X}_{n+k})^2)$.
- ▶ We know that the optimal estimator is $E(X_{t+k}|\mathcal{F}_n)$ to minimize mean square error. This is straightforward when we have a structured model like AR, MA or ARMA. In general the conditional expectation may not be computable, and in that case we fall back onto linear estimation.

Forecasting: 5

- ▶ In practice, we use the optimum linear forecast, i.e.

$$\hat{X}_{n+k} = a_0 + a_1X_1 + \cdots + a_nX_n.$$

- ▶ As n changes, the coefficients may change, so we actually need to represent this forecast equation as

$$\hat{X}_{n+k} = a_{0n} + a_{1n}X_1 + \cdots + a_{nn}X_n.$$

- ▶ So there are two issues to address: 1) for a fixed n , find the optimal coefficients $\mathbf{a}_n = (a_{0n}, \cdots, a_{nn})$ and 2) if we have $\mathbf{a}_n$ and the next value in the series, X_{n+1} , how can we update $\mathbf{a}_n$ to $\mathbf{a}_{n+1}$?

Forecasting: 6

Finding the best linear predictor is very similar to finding the optimal control variable in Monte Carlo simulation. We wish to minimize

$$S(\mathbf{a}) = E((X_{n+k} - (a_0 + a_1X_n + \cdots + a_nX_1))^2),$$

over $\mathbf{a} = (a_0, a_1, \cdots, a_n)$.

- ▶ We differentiate $S(\mathbf{a})$ sequentially with respect to the components of $\mathbf{a}$, obtain a system of linear equations to solve and check that the solution is a minimum.
- ▶ In this case one can simplify the calculation by interchanging the order of expectation and differentiation, i.e.

$$\frac{\partial}{\partial \mathbf{a}} E(S(\mathbf{a})) = E\left(\frac{\partial}{\partial \mathbf{a}} ((X_{n+k} - (a_0 + a_1X_1 + \cdots + a_nX_n))^2)\right),$$

Forecasting: 7

For a_0 we have

$$\begin{aligned} 0 = \frac{\partial}{\partial a_0} E(S(\mathbf{a})) &= E\left(\frac{\partial}{\partial a_0} (X_{n+k} - a_0 - \sum_{j=1}^n a_j X_{n+1-j})^2\right) \\ &= -2E(X_{n+k} - a_0 - a_1 X_n - \cdots - a_n X_1) \\ &= -2(\mu - a_0 - \mu(a_1 + \cdots + a_n)) \\ &= -2(a_0 + \mu(1 - (a_1 + \cdots + a_n))). \end{aligned}$$

This is easily solved

$$\hat{a}_0 = \mu(1 - \sum_{j=1}^n a_j).$$

Forecasting: 8

Differentiation with respect to a_i , $1 \leq i \leq n$ gives

$$0 = \frac{\partial}{\partial a_i} = -2E((X_{n+k} - a_0 - \sum_{j=1}^n a_j X_{n+1-j})X_{n+1-i}).$$

Equivalently,

$$E(X_{n+k}X_{n+1-i} - a_0X_{n+1-i} - a_1X_nX_{n+1-i} - \cdots - a_nX_1X_{n+1-i}) = 0,$$

or

$$\gamma(k+i-1) + \mu^2 - a_0\mu - a_1(\gamma(1-i) + \mu^2) - \cdots - a_n(\gamma(n-i) + \mu^2) = 0.$$

Forecasting: 9

This can be rewritten as

$$\gamma(k+i-1) = a_0\mu - \mu^2 + \sum_{j=1}^n a_j\mu^2 + \sum_{j=1}^n a_j\gamma(j-i),$$

and note $a_0\mu - \mu^2 + \sum_{j=1}^n a_j\mu^2 = 0$. This results in a set of linear equations relating the a_i coefficients in terms of the $\gamma(h)$ coefficients:

$$\mathbf{\Gamma}_n \mathbf{a}_n = \gamma_n(n-1),$$

where $\mathbf{\Gamma}_n = (\gamma(i-j))_{i,j=1,\dots,n}$, $\gamma_n(n-1) = (\gamma(n-1), \dots, \gamma(0))$. These are the **prediction equations**. Here $\mathbf{\Gamma}_n$ is the same matrix that arose in the general Yule-Walker equations, and we can find the coefficients of the best predictor as

$$\mathbf{a}_n = \mathbf{\Gamma}_n^{-1} \gamma_n(k).$$

Forecasting: 10

The best linear predictor is

$$\hat{X}_{n+k} = \mu + \hat{a}_1(X_n - \mu) + \cdots + \hat{a}_n(X_1 - \mu),$$

and its mean square prediction error is

$$E((X_{n+k} - \hat{X}_{n+k})^2) = \gamma(0) - \mathbf{a}_n^T \boldsymbol{\gamma}_n(k).$$

Thus linear prediction for a weakly stationary time series model is straightforward:

- ▶ Solve the prediction equations using the autocovariance function of the process
- ▶ Write down the expression for the best linear predictor
- ▶ Evaluate its mean squared error.

Forecasting: 11

Predicting a white noise process, $WN(\mu, \sigma_\epsilon^2)$

- ▶ Here $\gamma(h) = \sigma_\epsilon^2$ if $h = 0$ and 0 if $h \neq 0$. Thus $\Gamma_n = \sigma_\epsilon^2 \mathbf{I}_n$.
- ▶ Thus the prediction equations become $\mathbf{a} = \gamma_n = \mathbf{0}$.
- ▶ Thus the best linear predictor of a white noise process with mean μ at a future point in time, ϵ_{n+k} , is given by $\hat{\mu} + 0 = \hat{\mu}$. The mean square error is $E((\epsilon_{n+k} - \mu)^2) = \sigma_\epsilon^2$.
- ▶ There is no information in the data that can improve the minimum mean square error prediction

Forecasting: 12

We next predict an AR(1) process, beginning with predicting X_{n+1} from $X_1, \dots, X_n$, the one-step-ahead predictor.

- ▶ Recall for an AR(1) process, $\gamma(h) = \frac{\sigma_\epsilon^2 \phi_1^{|h|}}{1 - \phi_1^2}$
- ▶ We need to find $\mathbf{a}_n$ to satisfy the prediction equations: $\mathbf{\Gamma}_n \mathbf{a}_n = \gamma_n(1)$. Both $\mathbf{\Gamma}_n$ and $\gamma_n(1)$ have a common factor $\sigma_\epsilon^2 / (1 - \phi_1^2)$ that we omit in writing the equations.
- ▶ The first row of $\mathbf{\Gamma}$ is $(1, \phi_1, \dots, \phi_1^{n-1})$. Multiplying this by ϕ_1 gives exactly $\gamma_n(1)$. Consequently, if we set

$$\hat{\mathbf{a}}_n = (\phi_1, 0, \dots, 0),$$

then we have a solution to the prediction equations.

Forecasting: 13

- ▶ Consequently, the best linear one-step-ahead predictor of X_{n+1} given $\mathcal{F}_n$ is $\hat{X}_{n+1} = \hat{\mu} + \phi_1(X_n - \hat{\mu})$.
- ▶ The minimum mean square error is

$$\begin{aligned} E((X_{n+1} - \hat{X}_{n+1})^2) &= \gamma(0) - \mathbf{\hat{a}}\gamma_n(1) \\ &= \frac{\sigma_\epsilon^2}{1 - \phi_1^2}(1 - \phi_1^2) \\ &= \sigma_\epsilon^2 \end{aligned}$$

- ▶ This is a reduction from $\sigma_\epsilon^2/(1 - \phi_1^2)$.

Forecasting: 14

- ▶ Now we can predict further in time, e.g. $\hat{Y}_{n+k}$ for $k > 1$. The analysis is essentially identical, as all that changes is $\gamma_n(k) = \phi_1^k \gamma_n(1)$.
- ▶ The prediction equations are easily solved, we simply multiply $\mathbf{a}_n$ found before by $\phi_1^k, (\phi_1^k, 0, \dots, 0)$.
- ▶ The minimum mean square error for this k-step-ahead predictor is

$$\sigma_\epsilon^2 \frac{1 - \phi_1^{2k}}{1 - \phi_1^2}$$

- ▶ Note how the error increases as k increases, with the limit as $k \rightarrow \infty$ being the unconditional variance $\sigma_\epsilon^2 / (1 - \phi_1^2)$, an intuitive result.

Forecasting: 15

- ▶ Now the best linear predictor we derived for a weakly stationary time series (coming from solving the prediction equations) used the parameters of the time series as though we knew them.
- ▶ Obviously, in general we will not know those parameters (unless we generated the time series by simulation).
- ▶ We saw this problem in Simulation where we needed to use, say, a standard deviation of the data in defining the optimal control variable. In that case we used the data to estimate the standard deviation, resulting in a standard error (estimated standard deviation of an estimate).

Forecasting: 16

- ▶ This is the same situation. We have an optimal estimator (forecast) involving unknown parameters, so we substitute those for the unknown true values.
- ▶ However, the minimum mean squared error we computed does not reflect the added variability coming from the uncertainty in the estimated values, including σ_e .
- ▶ In addition to the additional uncertainty that arises from the estimation of the model parameter estimate, we should recognize that creating a “point” forecast is only part of the task.

Forecasting: 17

- ▶ We need to accompany the point forecast with a measure of uncertainty. Indeed, we really need to develop a **predictive distribution**.
- ▶ The predictive distribution is the probability distribution of our optimal forecast. We generally use the mean of that distribution if we must give a specific value, but the predictive distribution allows for the creation of confidence intervals for a prediction (“prediction intervals”).
- ▶ The generation of a predictive distribution can be done in theory for special favorable assumptions, or it can be done by Monte Carlo methods for more general assumptions.

Forecasting: 18

- ▶ Generally, with enough data, the model parameter estimates will be asymptotically normally distributed (e.g. MLE or moment estimators are so by the central limit theorem).
- ▶ If n is sufficiently large, the error from the parameter estimates likely dominates the error from the model and from the noise process in the model.
- ▶ In most cases, statistical software like statsmodels is used and will automatically compute forecasts and their limits.
- ▶ But for more careful work (for example, a concern for heavy tails), Monte Carlo simulation should be applied.

Forecasting: 19

- ▶ For example, given an AR(1) process, parameters ϕ_0, ϕ_1 and σ_ϵ^2 , a model for $\{\epsilon_t, -\infty < t < \infty\}$, and data $(X_1, \dots, X_n)$.
- ▶ Given this information, one can simulate many paths for the process using the AR(1) model, the parameter estimates and the assumptions about the noise process distribution.
- ▶ This would result in a set of values for X_{n+k} , and the distribution of these values gives the predictive distribution of the process. From that distribution, one can find a point prediction (usually the mean) and develop confidence limits using that distribution.

Forecasting Example: 1

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
start = '2011-01-01'
end = '2021-04-7'
stk = pdr.get_data_yahoo('ibm',start,end)
stk_adjclose = stk['Adj Close']
#More commands to get logret
#Order select ARMA Model (2,2)
model = sm.tsa.SARIMAX(logret, order = (2,0,2), trend = 'c')
res = model.fit()
model1= res.get_forecast(steps = 10)
print(model1_fcast.summary_frame())
```

Forecasting Example: 2

	coef	std err	[0.025	0.975]
intercept	0.0001	0.000	-0.001	0.001
ar.L1	-0.3246	0.057	-0.436	-0.214
ar.L2	0.1946	0.090	0.019	0.371
ma.L1	0.2762	0.057	0.165	0.388
ma.L2	-0.1794	0.096	-0.367	0.009
sigma2	0.0002	2.63e-06	0.000	0.000

Forecasting Example: 3

y	mean	mean_se	mean_ci_lower	mean_ci_upper
0	-0.000800	0.014499	-0.029216	0.027617
1	0.000654	0.014516	-0.027796	0.029104
2	-0.000266	0.014523	-0.028729	0.028198
3	0.000316	0.014525	-0.028153	0.028785
4	-0.000052	0.014526	-0.028523	0.028419
5	0.000181	0.014527	-0.028292	0.028653
6	0.000033	0.014527	-0.028439	0.028506
7	0.000126	0.014527	-0.028346	0.028599
8	0.000068	0.014527	-0.028405	0.028540
9	0.000105	0.014527	-0.028368	0.028578

Non-Stationary to Stationary

- ▶ Most series in financial economics and finance are not stationary. Since our methods are for stationary processes, we want to transform the original data set into one that is much closer to stationarity.
- ▶ There are several standard methods in common use to accomplish this including:
 - ▶ Note if the dataset exhibits unequal variance (other than volatility clustering), for example larger variance for higher values. [Consider a Box-Cox transformation.](#)
 - ▶ If the time series has a periodically repeating pattern, then remove a [seasonal component](#)
 - ▶ If the time series has a smooth upward or downward behavior, [fit and remove a trend component](#)
 - ▶ After removing trends and seasonal components, if the resulting series is not weakly stationary, consider [differencing](#)

Box-Cox Transformation

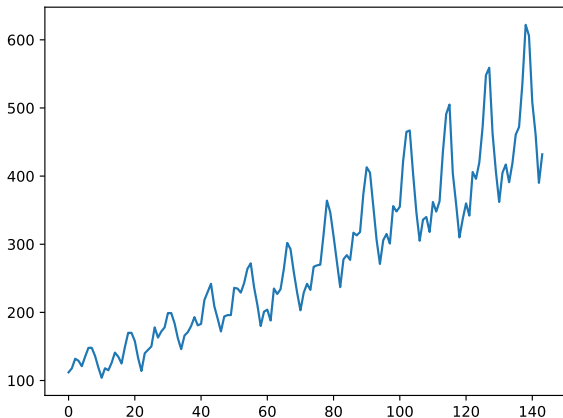
- ▶ Much of statistical analysis is done in the context of a symmetric error distribution. When there is a lack of symmetry, a Box-Cox or power transformation such as a square root or logarithmic transformation might help.
- ▶ The Box-Cox family is indexed by a parameter, λ mapping the data X into the transformed data X^λ , where

$$X^\lambda = \begin{cases} (X^\lambda - 1)/\lambda & \lambda \neq 0, \\ \log(X) & \lambda = 0. \end{cases}$$

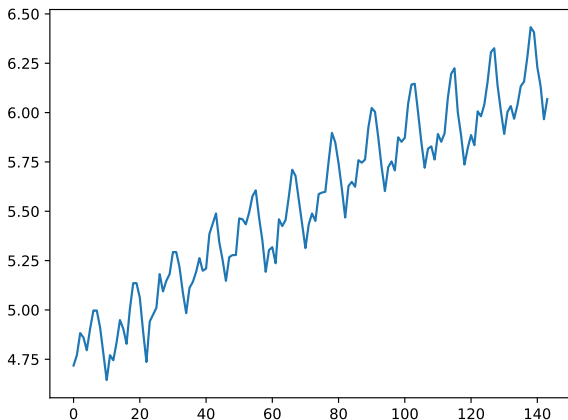
- ▶ The λ divisor is included to make the entire family continuous as $\lambda \rightarrow 0$, since

$$\lim_{\lambda \rightarrow 0} (X^\lambda - 1)/\lambda = \log(X).$$

International Airline Passengers, 1949-1960



Log(Int. Airline Passengers: 1949-1960)



Trend and Seasonal Components: 1

- ▶ After transforming to make the variances more equal, we must also consider trends and seasonality (periodicity).
- ▶ Let us first consider trends and let $X_t = m_t + Y_t$, where m_t is a trend and Y_t is weakly stationary.
- ▶ Differencing is a common way to handle the m_t term, but first we consider explicit modeling of m_t .
- ▶ One first decides on a parametric model for the trend, e.g.
 - ▶ Linear trend: $m_t = \beta_0 + \beta_1 t$,
 - ▶ Quadratic trend: $m_t = \beta_0 + \beta_1 t + \beta_2 t^2$
- ▶ One then estimates the coefficients using least squares.

Trend and Seasonal Components: 2

- ▶ An alternative approach is to use a **moving average smoother**. Choose a non-negative integer q , the half-bandwidth of the smoother). Then for $t = q + 1, \dots, n - q - 1$

$$\hat{m}_t = \frac{1}{2q + 1} \sum_{j=-q}^q X_{t+j}.$$

- ▶ Why does this work?

Trend and Seasonal Components: 3

$$\begin{aligned}\hat{m}_t &= \frac{1}{2q+1} \sum_{j=-q}^q X_{t+j} \\ &= \frac{1}{2q+1} \sum_{j=-q}^q (m_t + Y_{t+j}) \\ &= \frac{1}{2q+1} \sum_{j=-q}^q m_{t+j} + \frac{1}{2q+1} \sum_{j=-q}^q Y_{t+j} \\ &\approx m_t + \text{small error}\end{aligned}$$

Trend and Seasonal Components: 4

- ▶ The moving average smoother does preserve a linear trend. If $m_t = a_0 + a_1 t$:

$$\begin{aligned}\frac{1}{2q+1} \sum_{j=-q}^q m_{t+j} &= \frac{1}{2q+1} \sum_{j=-q}^q (a_0 + a_1(t+j)) \\ &= \frac{1}{2q+1} \sum_{j=-q}^q (a_0 + a_1 t) + \frac{1}{2q+1} \sum_{j=-q}^q a_1 j \\ &= a_0 + a_1 t + 0 \\ &= m_t\end{aligned}$$

- ▶ An alternative approach to the explicit modeling of a trend and seasonal (periodic) component is to difference the time series.

Trend and Seasonal Components: 5

- ▶ We next consider the case in which there is a seasonal (periodic) component with period d present.
- ▶ First, we define indicator functions,

$$I_j(t) == \begin{cases} 1 & \text{if } t \bmod d = j \\ 0 & \text{otherwise} \end{cases}$$

for $1 \leq j \leq d$.

- ▶ Second, we consider a model for the trend, for example a linear trend, $m_t = \beta_0 + \beta_1 t$. Thus our model becomes

$$\hat{X}_t = \beta_0 + \beta_1 t + \sum_{j=2}^d \beta_j I_j(t),$$

and we estimate the $d + 1$ parameters $\{\beta_i, i = 0, \dots, d\}$ using least squares by minimizing $\sum (X_t - \hat{X}_t)^2$.

Trend and Seasonal Components: 6

- Note that one of the indicators is omitted as the sum of all indicators must be 0. Define $\{\omega_j, 1 \leq j \leq d\}$ by $\omega_1 = 0, \omega_2 = \hat{\beta}_2, \dots, \omega_d = \hat{\beta}_d$. We need to adjust the seasonal components so that they sum to 0, consequently, we define $\bar{\omega} = \frac{1}{d} \sum_{j=2}^d \hat{\beta}_j$. The seasonal components are then given by $\hat{s}_j = \omega_j - \bar{\omega}, 1 \leq j \leq d$, with $\hat{s}_t = \hat{s}_{t-d}$ for $t > d$. The trend estimate is

$$\hat{m}_t = \hat{\beta}_0 + \hat{\beta}_1 t + \bar{\omega}.$$

The same ideas can be applied for more complicated trend models.

Differencing: 1

- ▶ We define the differencing operator ∇ by $\nabla X_t = X_t - X_{t-1} = (1 - B)X_t$. Note that one can create polynomials in ∇ and B for example:

$$\nabla^2 X_t = (1 - B)^2 X_t = (1 - 2B + B^2)X_t = X_t - 2X_{t-1} + X_{t-2}.$$

- ▶ Differencing can remove polynomial trends. Consider the linear and quadratic cases. Suppose $m_t = \beta_0 + \beta_1 t$, that is $X_t = \beta_0 + \beta_1 t + Y_t$, where Y_t is stationary.

$$\begin{aligned}\nabla X_t &= X_t - X_{t-1} \\ &= \beta_0 + \beta_1 t + Y_t - \beta_0 - \beta_1(t-1) - Y_{t-1} \\ &= \beta_1 + Y_t - Y_{t-1} \\ &= \beta_1 + \text{a stationary process}\end{aligned}$$

Differencing: 2

Similarly, suppose instead there is a quadratic term, i.e.

$X_t = \beta_0 + \beta_1 t + \beta_2 t^2 + Y_t$. Applying differencing twice, $(1 - B)^2 = 1 - 2B + B^2$, to the time series, we find

$$\begin{aligned}(1 - B)^2 X_t &= X_t - 2X_{t-1} + X_{t-2} \\&= \beta_0 + \beta_1 t + \beta_2 t^2 + Y_t \\&\quad - 2(\beta_0 + \beta_1(t-1) + \beta_2(t-1)^2 + Y_{t-1}) \\&\quad + \beta_0 + \beta_1(t-2) + \beta_2(t-2)^2 + Y_{t-2} \\&= 2\beta_2 + (Y_t - 2Y_{t-1} + Y_{t-2}) \\&= 2\beta_2 + \text{a stationary process}\end{aligned}$$

Differencing: 3

- ▶ One can also use differencing to remove both a seasonal and a linear trend. Suppose that $X_t = \beta_0 + \beta_1 t + s_t + \epsilon_t$, and the seasonal term has period d , i.e. $s_t = s_{t-d}$.
- ▶ In this case the differencing should be done over d time periods rather than simply one time period using ∇ . Instead we define a “d-lagged difference operator”

$$\nabla_d = X_t - X_{t-d} = (1 - B^d) \neq \nabla^d.$$

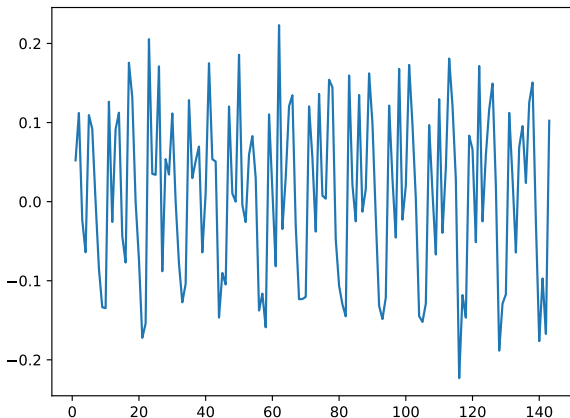
Differencing: 4

Applying this operator to the time series, we find

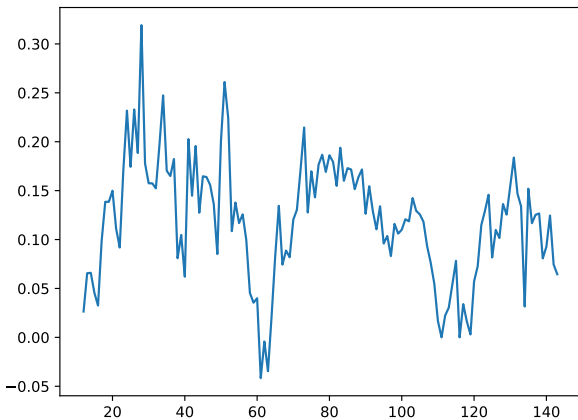
$$\begin{aligned}\nabla_d X_t &= \nabla_d m_t + \nabla_d S_t + \nabla_d \epsilon_t \\ &= m_t - m_{t-d} + S_t - S_{t-d} + \epsilon_t - \epsilon_{t-d} \\ &= \beta_1(t - (t - d)) + 0 + \epsilon_t - \epsilon_{t-d} \\ &= \beta_1 d + \text{a stationary process.}\end{aligned}$$

Thus this differencing operator has transformed the original time series into a weakly stationary process.

Difference(Log(Airline Passengers))



Seasonal Adj. (12) Log(Airline)



Diff(Seasonal12Log(Airline))

